



Chandigarh Engineering College Jhanjeri  
Mohali-140307  
Department of Artificial Intelligence (AI) and Data Sciences

# Smart Expense Tracker

**BACHELOR OF TECHNOLOGY**  
(Artificial Intelligence and Data Science.)



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## **DECLARATION**

We, Aditi and **Annu**, hereby declare that this project report entitled "**Smart Expense Tracker**" is an authentic record of our own work carried out under the training at **CGC University, Mohali**, in partial fulfillment of the requirements for the award of the **Bachelor of Technology (B.Tech.)** degree in **Computer Science & Engineering**.

We express our sincere gratitude to our esteemed guide, Mr. Chaitanya Singla , for their invaluable guidance, constant encouragement, and insightful suggestions throughout the course of this project. Their expertise and unwavering support were instrumental in the successful completion of this endeavour. We are also thankful to the Head of the Department, Ms. Sugandhi Ma'am for providing us with the necessary resources and conducive environment to carry out our research.

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**Signature of the Head of Department**  
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## Table of Contents

| S.No. | Contents                     | Page No |
|-------|------------------------------|---------|
| 1.    | Introduction                 | 5       |
| 2.    | System Requirements Analysis | 6       |
| 3.    | Implementation               | 7-8     |
| 4.    | Results and Discussions      | 9-10    |
| 5.    | Conclusion & Future Scope    | 11      |
| 6.    | Reference                    | 12      |



# Chapter 1

## Introduction

### 1.1 Project Introduction

The **Smart Expense Tracker** is a web-based application designed to help users manage and monitor their daily expenses efficiently. It provides a simple and user-friendly interface where users can add, view, search, and delete expense records while organizing them into different categories such as Food, Travel, Shopping, Bills, and Others. The application also generates category-wise expense reports and displays spending patterns using a pie chart, allowing users to better understand and manage their financial habits. Developed using **Python, Flask, SQLite, HTML, CSS, Bootstrap, and Matplotlib**, the system reduces the need for manual expense tracking, improves accuracy, and provides a secure platform for maintaining personal expense records.

### 1.2 Objectives of the Project

- To simplify the process of recording daily expenses.
- To provide an organized system for managing personal expense records.
- To allow users to add, view, search, and delete expense entries.
- To categorize expenses for better financial analysis.
- To generate category-wise expense reports.
- To visualize spending patterns using pie charts.
- To store expense data securely using SQLite.
- To develop a responsive, user-friendly, and easy-to-use web application.

### 1.3 Tools Learned

- **Python** – Backend programming language used to build the application logic.
- **Flask** – Lightweight Python web framework for developing the web application.
- **SQLite** – Database used to store expense records.
- **HTML** – Used to create the structure of web pages.
- **CSS** – Used for styling and designing responsive web pages.
- **Bootstrap** – Frontend framework used to create a modern and responsive user interface.
- **Matplotlib** – Python library used to generate pie charts for expense visualization.
- **Visual Studio Code (VS Code)** – Source code editor used for development.
- **Git & GitHub** – Version control and project management tools used to maintain the project source code.



## Chapter 2

# Software Requirement Analysis

### 2.1 Problem Definition

Traditional expense management is often carried out using notebooks, paper receipts, or manual calculations. These methods are time-consuming and prone to errors, making it difficult to track daily spending and analyze financial habits. Users may forget to record expenses, lose receipts, or find it challenging to calculate category-wise spending manually. Additionally, there is no easy way to search past expenses or visualize spending patterns, which can lead to poor financial planning and budgeting.

The **Smart Expense Tracker** addresses these problems by providing a simple web-based platform where users can digitally record, organize, search, and manage their daily expenses. The application also generates category-wise reports and displays spending distribution through a pie chart, making expense management more efficient and organized.

### 2.2 Functional Requirement

- Add new expense records with amount, category, description, and date.
- View all stored expenses in a structured table.
- Delete unwanted expense records.
- Search expenses by category, date, or description.
- Generate category-wise expense reports.
- Display expense distribution using a pie chart.

### 2.3 Non-Functional Requirements

- **Performance:** Processes expense records quickly with minimal response time.
- **Reliability:** Ensures accurate storage and retrieval of expense data.
- **Security:** Protects stored data and prevents accidental data loss.
- **Usability:** Provides a clean, simple, and beginner-friendly user interface.
- **Scalability:** Can be extended with features such as user authentication, budget tracking, monthly reports, and cloud database integration.
- **Maintainability:** Uses a modular project structure that allows easy updates, debugging, and future enhancements.

### 2.4 Advantages of the Proposed System

The proposed **Smart Expense Tracker** offers several advantages over traditional manual expense management:

- Eliminates the need for paper-based expense records.
- Reduces calculation and recording errors.
- Stores all expense information in a secure SQLite database.
- Enables quick searching of expense records.



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- Organizes expenses into different categories for better analysis.
- Displays spending patterns through a graphical pie chart.
- Saves time by automating expense management.
- Provides a centralized platform for tracking and managing personal expenses.
- Helps users understand and improve their spending habits.



## Chapter 3

# Implementation

### 3.1 System Implementation

The **Smart Expense Tracker** is implemented as a web-based application using the **Flask** framework in Python. The application follows a modular approach, where different functionalities such as expense management, report generation, chart visualization, and database operations are organized into separate modules. This structure makes the application easier to develop, maintain, and enhance.

### 3.2 Expense Management Module

The Expense Management Module is the core functionality of the Smart Expense Tracker. It enables users to record and manage their daily expenses through a simple and user-friendly interface.

#### Add Expense

Users can add a new expense by entering the following details:

- Amount
- Category
- Description
- Date

Before storing the expense, the system validates the entered information to ensure that all required fields are completed correctly. The validated data is then saved securely in the SQLite database.

#### View Expenses

Users can view all previously saved expense records in a well-organized table. The table displays the following information:

- Expense ID
- Date
- Category
- Description
- Amount

This allows users to easily keep track of their daily spending and manage their financial records efficiently.





### 3.3 Expense Viewing and Search Module

This module allows users to easily access and manage their stored expense records.

Users can:

- View all saved expenses.
- Search expenses by category.
- Search expenses by date.
- Search expenses by description.
- Display matching expense records instantly.

The module provides a simple and organized table view for better readability.

### 3.4 Category Report Module

The Category Report Module helps users analyze their spending habits by grouping expenses according to categories.

The system calculates:

- Total amount spent on Food.
- Total amount spent on Travel.
- Total amount spent on Shopping.
- Total amount spent on Bills.
- Total amount spent on Others.

This feature helps users understand where most of their money is being spent.

### 3.5 Database Integration

SQLite is used as the backend database for storing all expense records.

The database stores:

- Expense ID
- Amount
- Category
- Description
- Date

Flask establishes a secure connection with SQLite to perform Create, Read, Delete, and Search (CRUD) operations whenever required.

### 3.6 Data Visualization

The application includes a data visualization module using **Matplotlib**.

The system generates:



- Category-wise Pie Chart

The pie chart provides a graphical representation of the user's spending distribution across different expense categories, making financial analysis easier to understand.

### 3.7 Key Features Implemented

The following features have been successfully implemented in the Smart Expense Tracker:

- Add Expense
- View Expenses
- Delete Expense
- Search Expense
- Category-wise Report
- Pie Chart Visualization
- SQLite Database Integration
- Responsive User Interface

### 3.8 System Performance

The Smart Expense Tracker performs efficiently by automating the process of recording and managing daily expenses. It reduces manual effort, provides quick access to expense records, and enables users to analyze their spending patterns effectively.

The combination of **Flask** and **SQLite** ensures fast response time, lightweight deployment, and reliable performance for individual users and small-scale expense management applications. The application is easy to use, requires minimal system resources, and can be extended with additional features in the future.

The customer module provides all the essential functionalities required for managing laundry services. Customers can:

- Place new laundry orders.
- Select laundry service options.
- View previously placed orders.
- Track the current status of ongoing orders.
- Access their account securely.

The module is designed to provide a simple and convenient user experience.



## Chapter 4

# Results and Discussion

### 4.1 Results

The **Smart Expense Tracker** was successfully developed and tested as a web-based application. The system effectively helps users manage their daily expenses by allowing them to add, view, search, and delete expense records through a simple and user-friendly interface.

The application categorizes expenses and generates category-wise reports to help users analyze their spending habits. It also displays a pie chart that provides a clear graphical representation of expense distribution across different categories.

During testing, all major functionalities performed as expected. Expense management, searching, deleting records, report generation, and pie chart visualization were executed successfully without any significant errors. The integration of Flask with SQLite ensured smooth communication between the application and the database.

The developed system provides a reliable solution for organizing daily expenses, reducing manual record-keeping, and improving personal financial management.

### 4.2 Discussion

The implementation of the Smart Expense Tracker demonstrates how web technologies can simplify personal expense management. Compared to traditional methods such as notebooks or spreadsheets, the application offers better organization, improved accuracy, and quicker access to expense records.

Users can easily record their daily expenses, search previous records, and analyze spending based on different categories. The graphical pie chart makes it easier to understand spending patterns and identify areas where expenses can be reduced.

The modular structure of the application makes it easy to maintain and extend in the future. Additional features such as user authentication, budget tracking, monthly reports, cloud storage, and PDF or CSV export can be integrated without major modifications.

The project also demonstrates the effectiveness of using Flask and SQLite for developing lightweight, secure, and beginner-friendly web applications.

### 4.3 Benefits Achieved

The successful implementation of the Smart Expense Tracker provides several practical benefits:

- Eliminates manual expense record maintenance.
- Simplifies daily expense management.
- Stores expense records securely in a database.
- Provides quick searching of expense records.
- Displays category-wise expense reports.
- Visualizes spending patterns using pie charts.
- Reduces calculation errors and saves time.



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- Helps users make better financial decisions by analyzing spending habits.

#### 4.4 Limitations

Although the application performs efficiently, a few limitations still exist:

- The application does not include user registration or login functionality.
- Monthly expense reports are not currently available.
- Export to PDF or CSV is not implemented.
- Budget planning and spending alerts are not included.
- The application uses SQLite, which is suitable for small-scale projects but may require a more powerful database for larger applications.
- Internet connectivity is required when the application is deployed as a web application.

### Chapter Summary

The **Smart Expense Tracker** was successfully implemented and tested, demonstrating its ability to simplify personal expense management. The system provides efficient expense recording, searching, category-wise reporting, and graphical visualization using pie charts. Overall, the project achieves its objectives by reducing manual work, improving organization, and helping users better understand their spending habits while providing a strong foundation for future enhancements.



## Chapter 5

# Conclusion & Future Scopes

### 5.1 Conclusion

The **Smart Expense Tracker** was successfully designed and developed to provide a simple, efficient, and user-friendly solution for managing daily expenses. The application automates essential tasks such as adding, viewing, searching, deleting, and categorizing expense records through a centralized web-based platform.

The project successfully overcomes many of the challenges associated with traditional manual expense management, including paper-based record keeping, calculation errors, and difficulty in analyzing spending patterns. By storing all expense information in a centralized SQLite database, the system ensures data accuracy, reliability, and easy accessibility.

The implementation of **Flask** as the backend framework and **HTML, CSS, Bootstrap, and Jinja2** for the frontend resulted in a responsive and easy-to-use application. The integration of **Matplotlib** enables users to visualize their spending through category-wise pie charts, making expense analysis more effective.

Overall, the Smart Expense Tracker achieves its primary objective of simplifying personal expense management, reducing manual effort, improving financial organization, and helping users understand their spending habits. The project also demonstrates the practical application of Python web development technologies in solving real-world personal finance problems.

### 5.2 Future Scope

Although the current system fulfills its intended objectives, several enhancements can be incorporated in future versions to improve its functionality and usability.

The following improvements can be implemented:

- User registration and secure login system.
- Monthly and yearly expense reports.
- Budget planning and spending limit alerts.
- Export reports to PDF or CSV format.
- Email notifications for monthly expense summaries.
- Mobile application support for Android and iOS devices.
- Cloud database integration using MySQL or PostgreSQL.
- Multi-user support with individual expense records.
- Interactive dashboards with additional charts and analytics.



- AI-based expense prediction and personalized saving recommendations.

## Chapter 6

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